

NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC

Subject: Mechanical Engineering

Questions: 14

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AI-generated PYQs: 0

1. [UNKNOWN] paper-i

Q2. (a) The principal stresses at a point are given as per the following figure: $2\sqrt{25} \text{ N/mm}^2$ and 200 N/mm^2 . Figure Q 2(a) Determine the state of stress w.r.t. X and Y axes. Also determine the maximum shear stress and direct stress on the plane of maximum shear. 10 (b) Write down the equations for shear force and bending moment for various sections of the simply supported beam as shown in the figure. Also draw SFD and BMD. 10 10 kN 10 kN 5 kN/m Figure Q 2(b) In a single-acting four-stroke engine, the work done by the gases during the expansion stroke is 4 times the work done during the compression stroke. Work done during suction and exhaust stroke is negligible. The engine develops 18 kW at 240 rpm . The fluctuation of speed is limited to 1% of the mean speed on either side. The turning moment curve for compression and expansion may be assumed to be of triangular shape. Determine the MI of the flywheel. 10 Using Iron-Carbon phase diagram, illustrate the eutectoid portion and dividing point between steels and cast iron. Which elements make the steel as stainless steel? 10 FSI-P-MCHE 3 ===== PAGE 4 =====

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2. [UNKNOWN] paper-i

Q3. (a) A bar of 2 m length is rigidly fixed to a support at top section where diameter is 50 mm and remains constant up to a length of 1 m . For the remaining portion, the diameter is 25 mm . If a weight of 1000 N falls freely through 100 mm and lands uniformly on a rigid collar at the lowermost cross-section, calculate the stress and extension in the bar. Take $E = 21 \times 10^5 \text{ N/mm}^2$. 10 (i) What is an isochronous governor? Discuss the stability of the same. 4 (ii) Each arm of a Porter governor is 250 mm long. The upper arms are pivoted on the axis of the governor whereas the lower arms are pivoted at a distance of 20 mm from the axis. Each ball has a mass of 5 kg and the sleeve mass is 40 kg . The force of friction on the sleeve is 40 N . At the radius of rotation of balls of 140 mm , determine the range of speed for which the governor will not respond. 6 Calculate the number of atoms per unit cell of a metal having lattice parameter of 0.29 nm , density of 7.868 g/cc , atomic weight is 55.85 g/mol and Avogadro's number is 6.023×10^{23} . What is the crystal structure of metal? 10 With strength vs temperature diagram show applicability of general polymers, advanced polymers, steels, aluminium, super alloys, ceramics and ceramic matrix composites. Which material will be most suitable for a part facing temperature of 1200°C in aerospace vehicles? 10

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3. [UNKNOWN] paper-i

Q4. (a) In what ways are composites different from alloys? A composite is made of 40% carbon fibres by volume in epoxy matrix. The tensile strength of carbon fibres is 1650 MPa and of epoxy is 280 MPa . What will be the theoretical strength of composite? 10 (b) On the basis of structure, distinguish between glass, ceramics and glass-ceramics. What makes glass-ceramics an important engineering material? 10 FSI-P-MCHE 4 ===== PAGE 5 ===== (c) A simply supported beam of length L is loaded with load $w = w_0 \sin \frac{\pi x}{L}$ with x measured from the left support. Determine: 10 Equation representing the deflection of the beam Slope at the ends Maximum deflection (d) A four-crank engine has the two outer cranks set at 120° to each other and their reciprocating masses are each 40 kg . The distance between the planes of rotation of adjacent cranks are 400 mm , 600 mm and 600 mm . For a complete primary balance, determine the reciprocating masses and their relative angular position of each of the inner cranks. If the length of each crank is 250 mm and the length of each connecting rod is 1 m , calculate the maximum secondary unbalanced force for a speed of rotation of 300 rpm . 10 71X L' FSI-P-MCHE 5 ===== PAGE 6 ===== SECTION B

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4. [UNKNOWN] paper-i

Q5. (a) What are the advantages of laser beam machining over conventional machining? 8 During an electric discharge drilling of a 10 mm square hole in a low carbon steel plate of 5 mm thickness, brass tool and kerosene are used. The resistance and capacitance in the relaxation circuit are 50Ω and $10 \mu\text{F}$, respectively. The supply voltage is 200 volts

and the gap is maintained at such a value that the discharge (sparking) takes place at 150 volts. Estimate the time required to complete the drilling operation. 8 What is Material Requirement Planning (MRP) ? What are the major benefits of using MRP ? 8 The cost analysis of two products A and B is given below: Particulars Product A Product B
Material 7 2.50 per unit 25 45 Labour @ 7 1 per hour 12 Labour @ 7 1.50 per hour — 15 Variable overheads 2 5 Selling price 70 80 On the basis of above information, which product would you recommend to be manufactured if labour is key factor and if material is key factor? 8 Explain Static RAM, Dynamic RAM and Cache memory used in most computer systems. Also list the merits and demerits of Cache memory. 8

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5. [UNKNOWN] paper-i

Q6. (a) During an orthogonal machining operation on mild steel, the results obtained are uncut chip thickness, $t_1 = 0.25$ mm chip thickness, $t_2 = 0.75$ mm width, $w = 2.5$ mm rake angle, $\alpha = 00^\circ$ cutting force, $F_c = 950$ N thrust force, $F_T = 475$ N Determine the coefficient of friction between the tool and the chip. Determine the ultimate shear stress T_s of the work material. 15 F5I-P-MCHE 6 ===== PAGE 7 ===== 7 (b) Given below are values of y for certain values of x . Find the equation of regression line that describes the relation between x and y . Hence estimate the value of y when $x = 24$. 15

x	y
2	4
4	6
6	8
8	10
10	12
12	14
14	16
16	18
18	20
20	22
22	24
24	26
26	28
28	30
30	32
32	34
34	36
36	38
38	40
40	42
42	44
44	46
46	48
48	50

(c) (i) What do you understand by the term "tool life" ? 4 (ii) What is the relationship between cutting speed and tool life? 6

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6. [UNKNOWN] paper-i

Q7. (a) A company manufactures two products A and B. Product A requires 3 man hours and 1 machine hour. Product B requires 1 man hour and 4 machine hours. The total man hours available is 50. Total machine hours available are 40. If product A gives a profit of 20 per unit and product B gives a profit of 25 per unit, write down the constraints. How many products of each type should be produced to maximise profit? Also show the feasible region. 15 (b) A project is represented by the network shown below and has the following data: Task: A B C DE F GH I Optimistic time: 5 18 26 16 15 6 7 7 3 Pessimistic time: 10 22 40 20 25 12 12 9 5 Most likely time: 8 20 33 18 20 9 10 8 4 Determine the following: 15 Expected task times and their variance The critical path F5I-P-MCHE ===== PAGE 8 ===== (c) (i) What do you understand by Flexible Manufacturing System (FMS) ? 3 Which conditions are suitable for its application? 7

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7. [UNKNOWN] paper-i

Q8. (a) The production department of a company requires 3,600 kg of raw material for manufacturing a particular item per year. It has been estimated that cost of placing an order is 36 and the cost of carrying inventory is 25 percent of the investment in the inventories. The price of investment is 10 per kg. Determine the ordering policy for raw material (i) optimal lot size, (ii) optimal order cycle time and minimum yearly variable inventory cost. 15 (b) (i) Explain the functions served by jigs and fixtures. 7 What is an indexing jig? 3 Describe any two types of locators used in jigs. 5 (c) (i) Discuss various operators used in FORTRAN language and give a brief idea about the operator's precedence. 5 (ii) Write 'C' program to find the largest three elements from the given array and also check that array size must contain more than three elements. If array size is less than three elements then it must print that the array is invalid. 5 F5I-P-MCHE 8

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8. [UNKNOWN] paper-ii

Q2. (a) Prove that: $(aP)_{jk} = k(aP)_{jk}$ (5P), where $k \in \mathbb{C}$ $P \in \mathbb{C}^{n \times n}$ [Hint : $C_p = C_v \cdot a^T$] 13 (b) Three thin walled infinitely long hollow cylinders of radii 5 cm, 10 cm and 15 cm are arranged coaxially. The temperatures of the innermost and outermost cylinders are respectively 1000 K and 300 K. Calculate the steady state temperature of the middle cylinder surface and the heat flow per m² area of the innermost cylinder. Assume emissivity of all the cylinders to be 0.5 and vacuum in the spaces between the cylinders. Take $\epsilon = 5.67 \times 10^{-8}$ W/m²K⁴. 15 Explain, with the help of T-s diagram, that for atmospheric moist air $T_{\text{dry bulb}} > T_{\text{wet bulb}} > T_{\text{dew point}}$, in unsaturated condition, and $T_{\text{dry bulb}} = T_{\text{wet bulb}} = T_{\text{dew point}}$, in saturated condition. 10 F5I- D- MCHE 2 15 ===== PAGE 3 =====

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9. [UNKNOWN] paper-ii

Q3. (a) A refrigeration plant operates on Reversed Carnot cycle. The saturation temperatures in the condenser and evaporator are 40°C and -5°C respectively. The volumetric efficiencies of the compressor and expander are 100 percent each. Calculate (i) the refrigeration effect per kg of refrigerant, (ii) the coefficient of performance, (iii) the compressor displacement per kW of refrigeration effect, and (iv) the net work input per kg of refrigerant. Also, calculate the corresponding for a simple vapour compression refrigeration cycle. Assume no superheating either at the inlet or exit of the compressor, and no subcooling. Show the cycles on T-s diagram. Explain, why in practice a throttle valve is used in vapour compression refrigeration rather than an expander cylinder. The properties of the refrigerant are as follows . (The specific volume of the saturated liquid is negligible compared to vapour). Saturation Temperature $^{\circ}\text{C}$ Saturation Pressure bar Enthalpy, kJ/kg Entropy, kJ/kg K Specific Volume (m^3/kg) v_g h_f h_g s_f s_g 40 9.6065 238.535 36715 11298 1.5405 1817 (-5) 2.6096 $195-395$ $349-32$ 0.9831 $1-5571$ $64-963$ f and g refer to saturated liquid and saturated vapour respectively. 15 (b) A piston-cylinder device contains 0.8 kg of steam at 300°C and 1 MPa. Steam is cooled at constant pressure until one-half of the mass condenses. 15 Show the process on a T — V diagram. Find the final temperature. Determine the volume change. Determine the heat transfer. Given: Tables P Tsat v_f v_g h_f h_g (kPa) ($^{\circ}\text{C}$) (m^3/kg) (m^3/kg) (kJ/kg) (kJ/kg) 1.000 $6-97$ 0.00100 129.19 $29-303$ $2513-7$ $100-0$ 99.61 0.00104 $F6941$ 417.51 2675.0 1000.0 179.88 ft 001127 $0-19436$ 762.51 27771 FSI- D- MCHE 3 ===== PAGE 4 ===== Property: At $P = 1.00$ MPa and $T = 300^{\circ}\text{C} = 74246$ kJ/kg-K = $0-25799$ m^3/kg h 3051.6 kJ/kg (c) Water at 20°C flowing at the rate of 0.015 kg/s enters a 25 mm inner diameter tube which is maintained at a temperature of 90°C . Assuming hydrodynamically and thermally fully developed flow, determine the heat transfer coefficient and the tube length required to heat the water to 70°C . Given water properties at 20°C ; $\rho = 1000.5$ kg/ m^3 , $C_p = 4181-8$ J/kg K, Kinematic viscosity = $F006 \times 10^{-6}$ m^2/s . Properties of water at 45°C : $\rho = 992$ kg/ m^3 , $C_p = 4180$ J/kg K, $k = 0.638$ W/mK, Kinematic viscosity = 0.613×10^{-6} m^2/s . The average Nusselt number for the tube is 3.657 . 10

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10. [UNKNOWN] paper-ii

Q4. (a) In a parallel flow heat exchanger, engine oil enters a heat exchanger at 150°C and leaves at 80°C . The cooling water enters at 30°C and leaves at 65°C . If the fluid flow rates and the inlet conditions are unchanged, find exit temperature of each stream in counterflow exchanger. 15 (b) A building has the following calculated cooling loads : RSH gain = 310 kW, RLH gain = 100 kW. The building's inside space is maintained at the following conditions of Room DBT = 25°C and Room RH = 50% . The outdoor air is at 28°C DBT and 50% RH. And 10% by mass of air supplied to the building is outdoor air. If the air supplied to the space is not to be at a temperature lower than 18°C , find minimum amount of air supplied to the space in m^3/s , volume flow rate of air entering the coil, capacity of the cooling coil in TR, and ADP and bypass factor of the cooling coil. Also, draw the layout sketch. (Psychrometric chart is attached) 15 FSI- D- MCHE 4 ===== PAGE 5 ===== 17.0 035 140 145 $12,5$ 130 155 $.01$ 0 $.040$ $ciis$ n $-a$ $-$ $.19$ PSYCHROMETRIC CHART BAROMETRIC PRESSURE 1.01325 bat SEA LEVEL 0.030 ten 4 on tosen effn on 0.024 0.023 0.022 ILN1 0.45 '11 :Aar vow $411116..$ kidligr $a..1111$,ah,,a%fillfeik9AlinalleatataW: eia wArei P 91161411111141101 lr $11111173111..$ 0019 GAM CLA17 CLOM 0.015 0.014 0.013 2 1 -003 001 0 471 toatAgraufsdnisiwmatian 'MIMI Ira Ofref %V IWO srepniiistonstegt,min Satirei MVO $11.240.10$ SAAB IPArSiORININVIratiSiVi .. $\bullet 4155151$ " Mite Medi Stairlier -41~11110.11bh .11 Zia!: 0 -- -10 -6 0 0.75 m'Ag 60 30 95 40 0 15 . 20 25 DRY BULB TEMPERATURE. $^{\circ}\text{C}$ 0.80 m^3/kg BELOW 0°C PIIOPERTIES AND ENTHALPY DEVIATION LINES ARE FOR ICE 0.35 mifkg 0 90 malkg SPECIFIC VOL LME, m^3/kg DRY AIR ===== PAGE 6 ===== (c) In the early development of steam power plants, approximately 112 kg of coal is required per kilowatt hour. Assume that the mean temperature at which heat was supplied is 175°C and heat was rejected is 100°C at that time. Presently, assume that heat is supplied at a mean temperature of 380°C and rejected at 32°C . The ratio of the actual thermal efficiency to that of the Carnot cycle today is about 1.25 times that of earlier years. Assuming the same heating value for coal in both cases, calculate the amount of coal now required per kilowatt hour. 10 V FSI- D- MCHE 6 ===== PAGE 7 ===== SECTION B

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11. [UNKNOWN] paper-ii

Q5. (a) Compare the cooling effect of fuel evaporation on charge temperature in a turbocharged spark ignition engine for the following two cases : The carburettor is placed before the compressor The carburettor is placed after the compressor The specific heat capacity of the air and the latent heat of evaporation of the fuel are both constant. For the air/fuel ratio of $12.5 : 1$, the evaporation of the fuel causes a 25 K drop in mixture temperature. The compressor

efficiency is 70% for the pressure ratio of 1-5, and the ambient temperature is 15°C. Assume the following values : for air, $C_p = 0.71 \text{ kJ/kgK}$, $\gamma = 1.4$, for air/fuel mixture, $C_p = 0.85 \text{ kJ/kgK}$, $\gamma = 1.34$. Finally, compare the compressor work in both cases. 8 A Ford 2-stroke engine has a swept volume of 1.2 litres. At a speed of 5500 rpm, the full load torque is 108 Nm and the brake specific fuel consumption is 294 g/kWh. Calculate the brake mean effective pressure, and the brake thermal efficiency (assume a calorific value of 43 MJ/kg). Give at least 5 reasons why the brake thermal efficiency is lower than the value predicted by the Otto cycles analysis. 8 What is meant by "work done factor" in the axial flow compressors ? Explain how the work done factor value varies with the number of stages. Why is the work done factor not considered in centrifugal compressors ? 8 What is 'slip factor' for a centrifugal compressor stage ? What is its effect on the flow and the pressure ratio in the stage? 8 FSI- D- MCHE 7 ===== PAGE 8 ===== 50 m Generator --> 5000 kg/s Turbine (e) Water in a large lake is to be used to generate electricity by the installation of a hydraulic turbine-generator located, where the depth of water is 50 m (see Figure). Water is to be supplied at the rate of 5000 kg/s. The electric power generated is 1862 kW and the generator efficiency is 95%. Determine (i) the overall efficiency of the turbine generator, (ii) the mechanical efficiency of the turbine, and (iii) the shaft power supplied by the turbine to the generator. - Density of water is 1000 kg/m³ and gravitational acceleration is 9.81 m/s².

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12. [UNKNOWN] paper-ii

Q6. (a) Reciprocating internal combustion engines have been fitted with ingenious mechanisms that allow the expansion ratio (r_e) to be greater than the compression ratio (r_c). When such a system is modelled by an ideal gas cycle there is (i) heat addition at constant volume, (U) some heat rejection at constant volume, (iii) and some heat rejection at constant pressure, to complete the cycle. The process is shown in the following figure on a T—s plane: > Depict the processes on a P-V plane. The constant-volume temperature rise (2 3) is $0T_1$. Derive an expression for the cycle efficiency in terms of r_e , r_c and γ . State any assumptions. 15 8 FSI- D- MCHE ===== PAGE 9 ===== In an ideal gas turbine cycle with reheat, air at state (P_1 , T_1) is compressed to (r_{p1}). in the compressor and heated to T_{3x} in the combustion chamber. The air is then expanded in two stages, each turbine having the same pressure ratio, with reheat to T_3 between the stages. Assuming the working fluid to be a perfect gas with constant specific heats, and that the compression and expansion are isentropic, show that the specific work output will be a maximum when 'r', the compressor pressure ratio is given by $\gamma/2$ $r(Y - 1)/Y = (13$ where γ is the ratio of specific heats, C_p/c . Draw the schematic arrangement of the cycle and the corresponding T—s diagram also. 15 C12H26 is burned at constant volume with no excess air. The initial temperature is 30°C. Assuming complete combustion, determine the theoretical maximum temperature when there is no dissociation. Use the following enthalpy values (J/mole) of different substances for the calculations : 10 C121-126 75, 79, 383 CO₂ 1, 46, 203 1120 1, 18, 329 N₂ 88, 122

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13. [UNKNOWN] paper-ii

Q7. (a) In a power plant employing Rankine cycle with reheat, superheated steam at 150 bar and 500°C ($h = 3310.6 \text{ kJ/kg}$; $s = 6.3487 \text{ kJ/kg K}$) enters the first stage of the turbine. The condenser is maintained at 0.1 bar ($v_f = 0.001 \text{ m}^3/\text{kg}$, $h_f = 191.83 \text{ kJ/kg K}$, $h_g = 2584.8 \text{ kJ/kg}$, $s_f = 0.6493 \text{ kJ/kg K}$, $s_g = 8.1511 \text{ kJ/kg K}$). The exit steam from the first stage of turbine is reheated to 500°C before it is fed to the second stage of the turbine. Calculate the thermal efficiency of the power plant if steam expands to 90 bar in the first stage of the turbine. 60 bar in the first stage of the turbine. 15 FSI- D- MCHE 9 ===== PAGE 10 ===== From the superheated steam tables at 90 bar T (°C) h (kJ/kg) s (kJ/kg K) 400 3125.5 61067 500 3389.2 6.6728 From the superheated steam tables at 60 bar T (°C) h (kJ/kg) s (kJ/kg K) 300 2885.0 6-0692 400 31801 6.5462 500 3422.2 6.8818 A gas engine operating on methane (CH₄) at 1500 rpm, full throttle, generates the following emissions measured on a dry volumetric basis : CO₂ 10-4% CO 11% 112 0.6% O₂ 0.9% NO 600 ppm HC 1100 ppm (as methane) If the specific fuel consumption is 250 g/kWh, calculate the specific emissions of carbon monoxide, nitrogen oxide, and unburned hydrocarbons (that is g/kWh). Why is the specific basis more relevant than a percentage basis? 15 A model of a torpedo is tested in a water filled towing tank at a velocity of 244 m/s. The prototype is expected to attain a velocity of 6-1 m/s in water. What would be the model speed if tested in a wind tunnel filled with air at a pressure of 20.0 bar and constant temperature of 27°C ? The absolute viscosity of air may be assumed as $1.845 \times 10^{-5} \text{ N.s/m}^2$, the molecular weight of air as 28.97 kg/kmol and the universal gas constant as 8.315 kJ/kg-mol K. The kinematic viscosity of water may be taken as $1.13 \times 10^{-6} \text{ m}^2/\text{s}$. 10 FSI- D- MCHE 10 ===== PAGE 11 =====

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14. [UNKNOWN] paper-ii

Q8. (a) Starting with the differential form of the conservation equations, show that the flow velocity increases with heat addition in subsonic Rayleigh flow, but decreases in supersonic Rayleigh flow. Also, draw the T-s diagram. 15 (b) At the mean diameter of a gas turbine stage, the blade velocity is 350 m/s. The blade angles at the inlet and exit are 20° and 54° (respectively) with respect to axial direction. The blades at this section are designed to have a degree of reaction of 50 percent. The mean diameter of the blades is 0.432 m and the mean blade height is 0.07 m. Assuming that the whirl velocity varies inversely with respect to radius, estimate: the flow velocity, the angles of blades at the tip and at the root, and the degree of reaction at the tip and at the root of the blades. 15 (c) Briefly explain the stages of combustion in SI engines elaborating the flame front propagation. 10 FSI- D- MCHE 11

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